The Mathematics of Uncertainty : An Introduction to Fuzzy Sets

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Background: When Math Meets Ambiguity

The real world isn't crisp.

- Hot ↔ Cold
- Tall ↔ Short
- Near ↔ Far

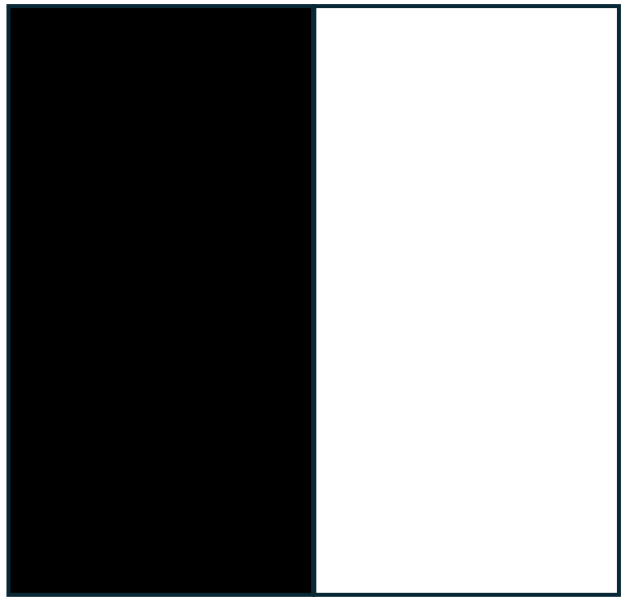
But classical mathematics avoids vagueness.

Zadeh (1965) wrote:

"Human reasoning is not always precise, but it is remarkably consistent."

Question: **Can we model ambiguity itself mathematically?**

The Idea of a Fuzzy Set



Fuzzy sets express **how much**, rather than **simply yes or no**.

The Difference between Randomness and Vagueness

- Randomness is modeled by probability theory.
- Vagueness is modeled by fuzzy sets and fuzzy logic.

Is a Fuzzy Set
Still a Set?

Informal Description of Sets

- A set is a collection of objects.

New Perspective on Sets

- View a set as **yes/no** test.
- For $X \subseteq A$, Define a classifier

$$f: A \to \{\mathbf{0}, \mathbf{1}\}, \qquad f(x) = \begin{cases} 1 & if\ x \in X \\ 0 & if\ x \notin X \end{cases}$$

Toward Fuzzy Sets

- Extend to

$$f: A \to [\mathbf{0}, \mathbf{1}]$$

- The value encodes the **degree of belonging**.

Definition of Fuzzy Sets

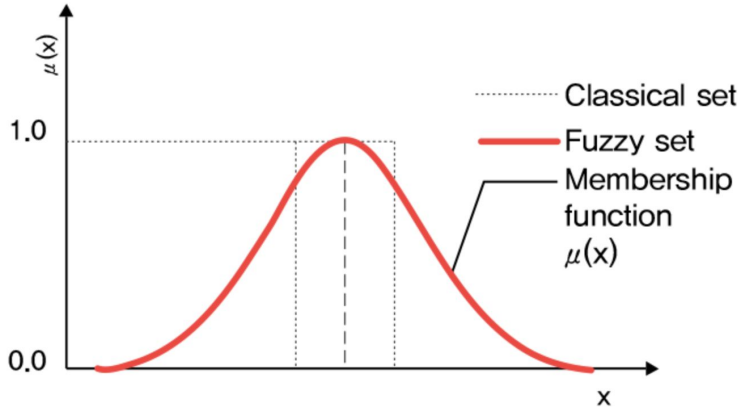
Definition. Let X be a set (universe).

A is called a **fuzzy subset** of X if A is **a set of ordered pairs**:

$$A = \{ \langle x, \mu_A(x) \rangle : x \in X \ \ \mu_A(x) \in [0,1] \}$$

where $\mu_A(x)$ is the grade(degree) of membership of x in A.

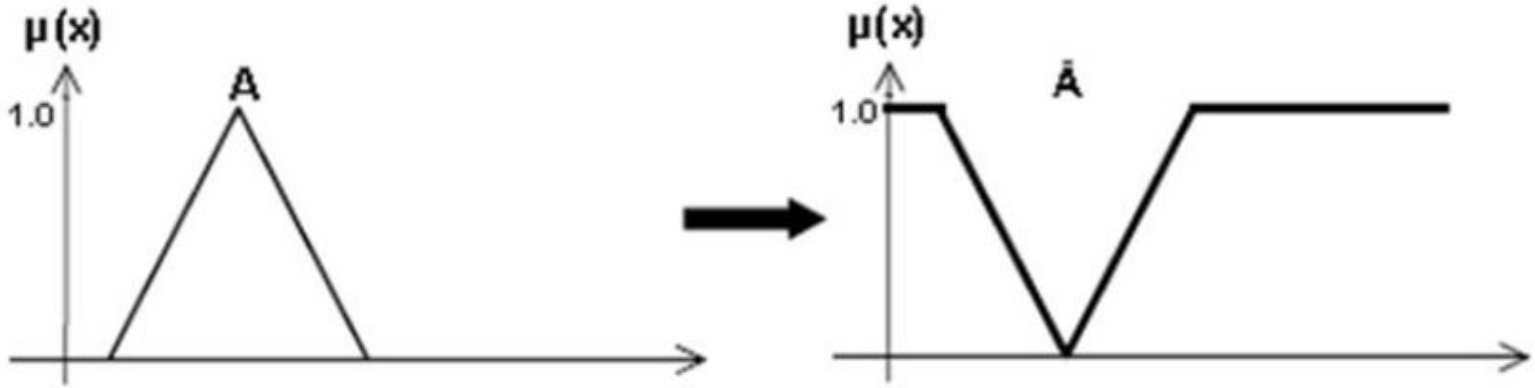
The function $\mu_A(x)$ is called the **membership function** of A.



Some Operations on Fuzzy Sets

Definition. The membership function of the complement $C = A^c$ of the fuzzy set A is defined by:

$$\mu_C(x) = 1 - \mu_A(x)$$



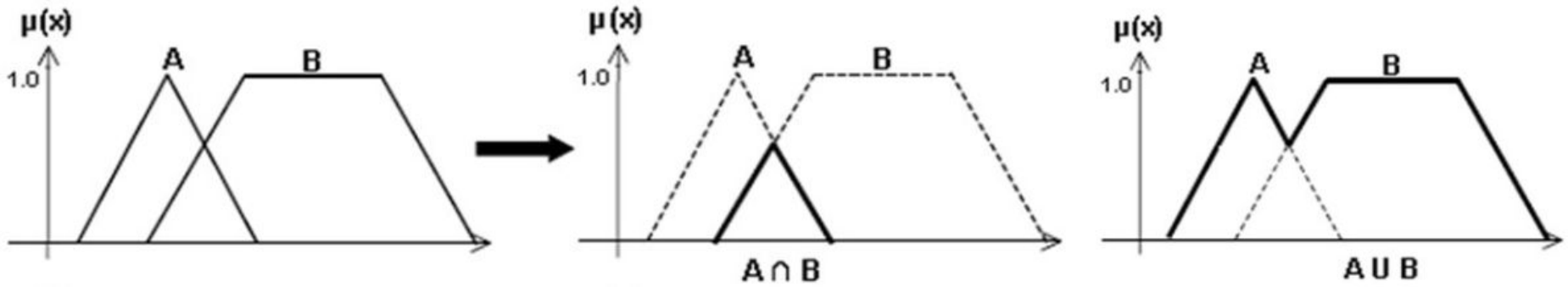
Some Operations on Fuzzy Sets

Definition. The membership function of the intersection $D = \boldsymbol{A} \cap \boldsymbol{B}$ of two fuzzy sets A and B is defined by:

$$\mu_D(x) = \mathbf{min}(\mu_A(x), \mu_B(x))$$

Definition. The membership function of the union $\mathrm{E} = \boldsymbol{A} \cup \boldsymbol{B}$ of two fuzzy sets A and B is defined by:

$$\mu_E(x) = \mathbf{max}(\mu_A(x), \mu_B(x))$$



(Generalized Conjunction: t-Norms)

- For intersection $D = \boldsymbol{A \cap B}$ of two fuzzy sets A and B,

$$\mu_D(x) = \mathbf{min}(\mu_A(x), \mu_B(x))$$

$$\Downarrow$$

Definition. A **t-norm**(triangular norm) is a bivariate function:

$$t\colon [0,1]\times[0,1] \to [0,1]$$

with the following properties:

1. $t(0,0) = 0$
2. $t(x,1) = x$
3. $t(u,v) \leq t(w,z)$ if $u \leq w$ and $v \leq z$ (Monotonicity)
4. $t(x,y) = t(y,x)$ (Symmetry)
5. $t(x,t(y,z)) = t(t(x,y),z)$ (Associativity)

$$\Downarrow$$

(Generalized Disjunction: t-Conorms)

- For union $\mathrm{E} = \boldsymbol{A} \cup \boldsymbol{B}$ of two fuzzy sets A and B,

$$\mu_E(x) = \boldsymbol{max}(\mu_A(x), \mu_B(x))$$

$$\Downarrow$$

Definition. A **t-conorm**(triangular norm) is a bivariate function:

$$c\colon [0,1]\times[0,1] \to [0,1]$$

with the following properties:

1. $c(1,1) = 1$
2. $c(x,0) = x$
3. $c(u,v) \leq c(w,z)$ if $u \leq w$ and $v \leq z$ (Monotonicity)
4. $c(x,y) = c(y,x)$ (Symmetry)
5. $c(x,c(y,z)) = c(c(x,y),z)$ (Associativity)

$$\Downarrow$$

Are Fuzzy Sets
Well Defined
within Classical Set Theory?

Axioms of a Set in Classical Set Theory

- Zermelo-Fraenkel Set Theory with the **Axiom of Choice** (**ZFC**)

- **Extensionality Axiom**
- **Empty Set Axiom**
- **Subset Axiom**
- **Pairing Axiom**
- **Union Axiom**
- **Power Set Axiom**
- **Infinity Axiom**
- **Replacement Axiom**
- **Regularity Axiom**

How Fuzzy Sets Fit into ZFC

- Let A, B be fuzzy subsets of X.

- **Extensionality Axiom**

$$A = B \leftrightarrow \forall x(x \in A \leftrightarrow x \in B)$$

$$\boldsymbol{A = B \leftrightarrow \forall x(\mu_A(x) = \mu_B(x))}$$

- **Subset Axiom** ($\varphi(x)$: a formula)

$$\exists S \forall x(x \in S \leftrightarrow (x \in A \wedge \varphi(x)))$$

$$\boldsymbol{\exists S \forall x \forall y(\langle x, y\rangle \in S \leftrightarrow (x \in X \wedge y \in [0,1] \wedge y = \mu_A(x) \wedge \varphi(x, y)))}$$

How Fuzzy Sets Fit into ZFC

- Let A be a set of fuzzy subsets of X.

- **Pairing Axiom**

$$\exists A \forall x (x \in A \leftrightarrow (x = u \vee x = v))$$

$\boldsymbol{u, u \rightarrow \{u\}}$ **and** $\boldsymbol{u, v \rightarrow \{u, v\}}$ **thus** $\boldsymbol{\{\{\mathrm{u}\}, \{\mathrm{u}, \mathrm{v}\}\} = \langle u, v \rangle}$ **exist.**

- **Union Axiom**

$$\exists U \forall x (x \in U \leftrightarrow \exists C (x \in C \wedge C \in A))$$

$$\boldsymbol{\exists U \forall x (\mu_U(x) = sup_{C \in A}\, \mu_C(x))}$$

Fuzzy Sets as Ordinary Sets

Definition. Let X be a set (universe).

A is called a **fuzzy subset** of X if A is **a set of ordered pairs**:

$$A = \{ \langle x, \mu_A(x) \rangle : x \in X \ \ \mu_A(x) \in [0,1] \}$$

where $\mu_A(x)$ is the grade(degree) of membership of x in A.

The function $\mu_A(x)$ is called the **membership function** of A.

- In ZFC:
  - X is a set, $[0,1]$ is a set.
  - $X \times [0,1]$ is a set. (Cartesian product)
  - Any subset $A \subseteq X \times [0,1]$ is again a set.

Choosing the Universe X

- In fuzzy set theory, the **universe X** is always a **crisp classical set.**

- We decide what X is:
  - Temperatures: $X = \mathbb{R}$
  - Image pixels: $X = \{1, \cdots, M\} \times \{1, \cdots, M\}$

- Once X is fixed as a set,
  every fuzzy set on X is just a subset of $X \times [0,1]$.

Fuzzy Sets

Fit Perfectly

within Classical Set Theory!

Where Do Fuzzy Sets Make a **Real Difference**?

Fuzzy Control:
Smooth Action from Vague Rules

1. Measure input x.
2. Compute how much x belongs to each fuzzy set.
3. Activate fuzzy rules with those membership strengths.
4. Combine all rule outputs into a single fuzzy action.
5. Defuzzify(e.g., center of gravity) to precise output y.

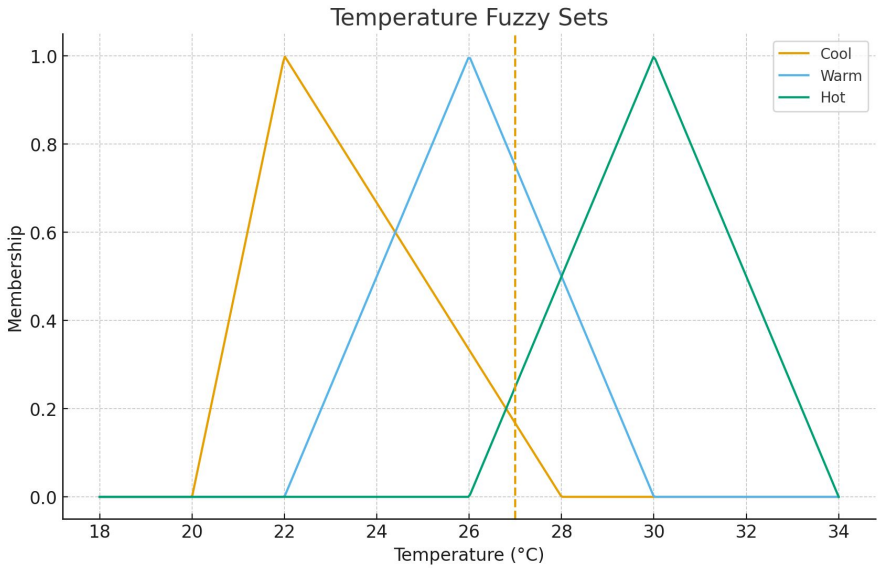
Example of Fuzzy Control: Air Conditioning

1. Measure current temperature T.

$$\boldsymbol{T = 27°\mathrm{C}}$$

2. Compute how much T belongs to each fuzzy set. (Cool/Warm/Hot)

$$\boldsymbol{\mu_{Cool}(27°\mathrm{C}) = 0.17}, \qquad \boldsymbol{\mu_{Warm}(27°\mathrm{C}) = 0.75}, \qquad \boldsymbol{\mu_{Hot}(27°\mathrm{C}) = 0.25}$$



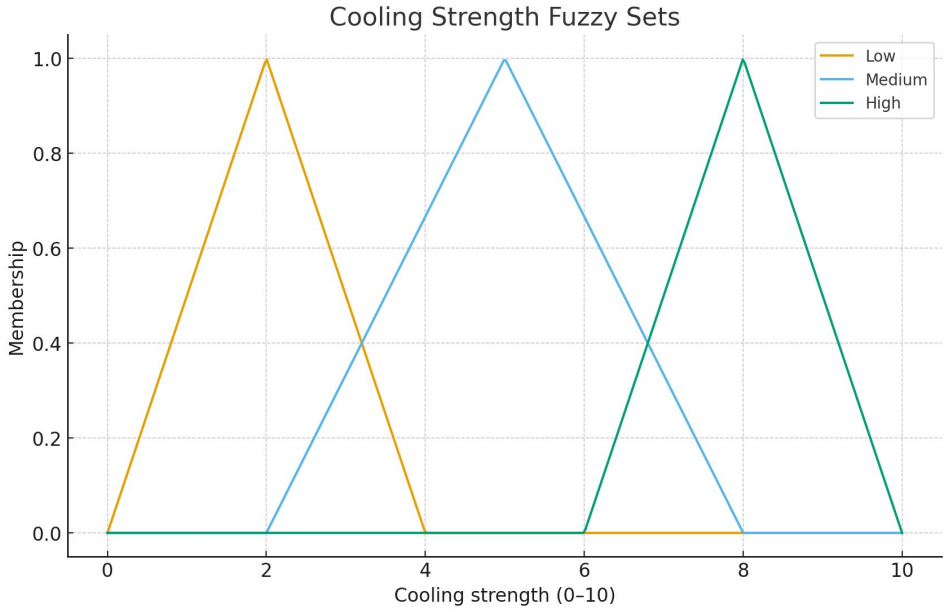
Example of Fuzzy Control: Air Conditioning

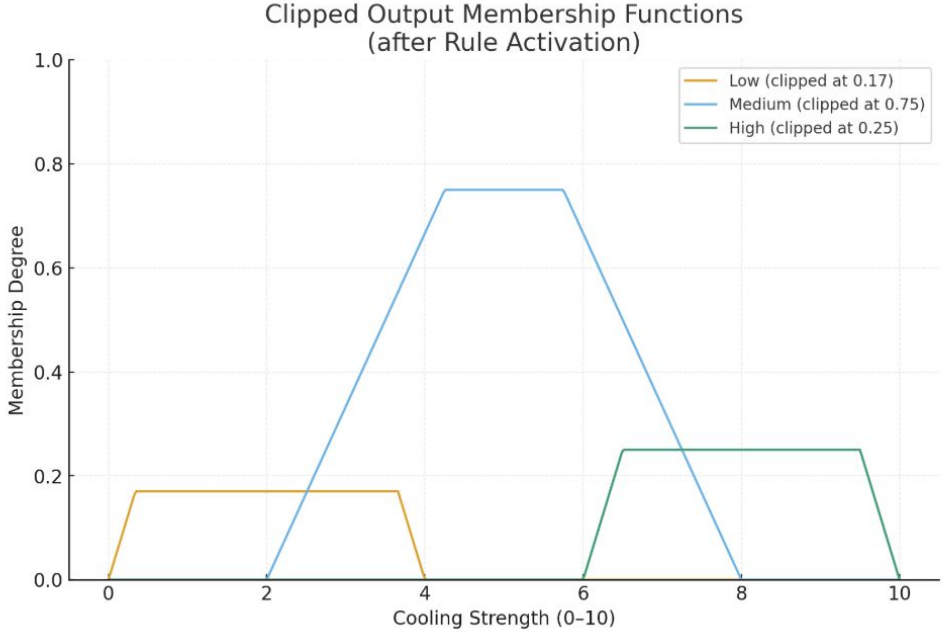
3. Activate fuzzy rules with those membership strengths.

If Cool, then Low Cooling.
If Warm, then Medium Cooling.
If Hot, then High Cooling.

activation level →

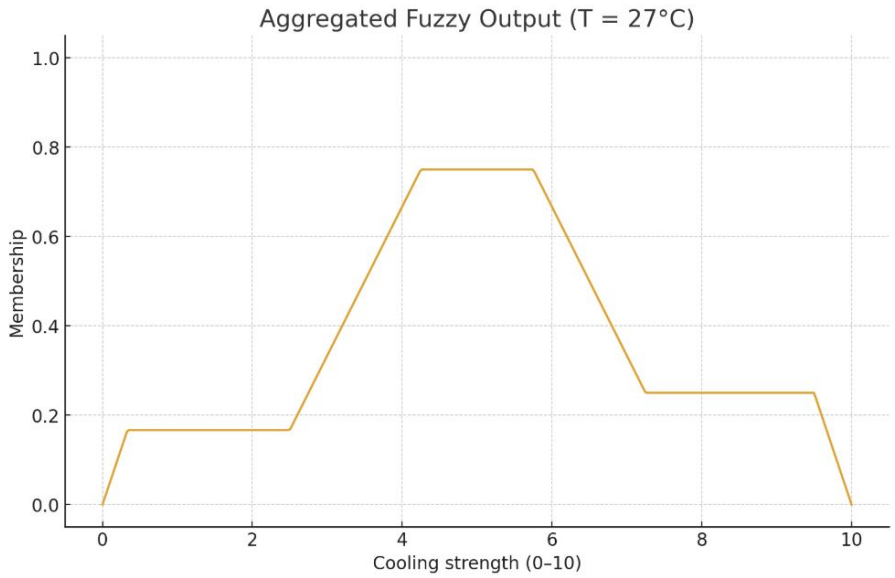
0.17
0.75
0.25





Example of Fuzzy Control: Air Conditioning

4. Combine all rule outputs into a single fuzzy action. (**Aggregation)**



5. Defuzzify(e.g., center of gravity) to precise cooling level. (**Defuzzification**)

Cooling Strength ≈ 5.2

Other Applications

- Home appliances (washing machines, air conditioners, rice cookers)
- Automotive (automatic braking, transmission control)
- Robotics (smooth motion, adaptive grasping)
- Finance (risk scoring with vague indicators)
- Medicine (diagnosis support with fuzzy symptoms)
- Image processing (edge detection, segmentation)
- Decision-making systems (expert systems, recommender logic)

Q. Why Do We Need Fuzzy Sets?

A. "Quantify Vagueness"
→ Go near to real-world problem!